

RACING TIRE CHARACTERIZATION AND PERFORMANCE MODELLING

Technical Report

*Analysis, Characterization and Engineering Interpretation of
Experimental Tire Test Data.*

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Executive Summary

Background:

The tire is the only component of a vehicle that remains in direct contact with the road surface and is therefore responsible for transmitting all forces required for acceleration, braking, and cornering. Despite its seemingly simple appearance, tire behaviour is highly complex and influenced by several interacting parameters, including vertical load, slip angle, camber angle, inflation pressure, temperature, and surface conditions. The nonlinear relationship between these variables makes tire analysis one of the most important areas of vehicle dynamics engineering.

Modern tire testing facilities generate large volumes of high-quality experimental data. However, raw data alone provides little engineering value unless it is systematically processed, visualized, and interpreted. Meaningful insights can only be obtained by understanding the physical behaviour represented by the data and relating it to real-world vehicle performance. For this reason, tire characterization forms a critical step in the development of accurate vehicle dynamics models and performance simulations.

Aim:

The primary aim of this project was to transform raw experimental tire test data into meaningful engineering information through a structured process of data analysis and interpretation. Rather than focusing solely on numerical results, the objective was to develop a deeper understanding of how a racing tire behaves under different operating conditions and how these behaviours influence vehicle performance.

The study focused on investigating key tire characteristics including lateral force generation, load sensitivity, camber sensitivity, cornering stiffness, and self-aligning moment behaviour. By extracting and interpreting these characteristics, the project aims to establish a strong foundation for future tire model development and a broader understanding of racing vehicle dynamics.

Methodology:

The analysis began with an exploration of the dataset to understand the test conditions and overall data structure. Key parameters such as slip angle, vertical load, and camber angle were examined, while histograms and preliminary plots were used to identify dominant load bands and assess data quality.

The dataset was then filtered into representative load ranges to study tire behaviour under controlled conditions. Characteristic tire curves were generated, and important performance parameters such as peak lateral force, friction coefficient, slip angle, and camber angle were extracted. Further investigations were carried out to evaluate load sensitivity, camber sensitivity, cornering stiffness, and self-aligning moment behaviour.

Throughout the project, the focus was not only on extracting numerical results but also on understanding the physical behaviour of the tire and relating the observed trends to real-world vehicle dynamics.

Results:

The tire exhibited the characteristic nonlinear behaviour expected of a racing tire. Lateral force increased approximately linearly with slip angle at small slip angles before entering a transitional region and eventually reaching saturation at higher slip angles. Peak lateral force generation was successfully identified, along with the corresponding operating conditions of vertical load, slip angle, and camber angle.

Load sensitivity analysis demonstrated that increasing vertical load increased the absolute lateral force generated by the tire. However, the increase in force was not proportional to the increase in load, resulting in a reduction in the effective friction coefficient at higher loads. This behaviour confirmed the presence of tire load sensitivity and highlighted the importance of managing weight transfer in high-performance vehicles.

The investigation also showed that cornering stiffness increased with vertical load, indicating a more responsive tire under higher loading conditions. Analysis of self-aligning moment behaviour revealed that maximum steering feedback occurred before the tire reached maximum lateral force, illustrating the reduction in pneumatic trail as the tire approached saturation. These findings provided valuable insight into both tire performance and driver feedback characteristics.

Conclusions:

Beyond the numerical results, the project provided a deeper understanding of the physical mechanisms responsible for tire force generation and their influence on vehicle behaviour. The findings reinforced key vehicle dynamics concepts such as the nonlinear nature of tire performance, the effects of load transfer, and the relationship between steering feedback and available grip.

The methodology developed throughout this work provides a repeatable framework for future tire investigations and establishes a strong foundation for the development of more advanced tire models. Furthermore, the project represents an important step in converting raw experimental data into practical engineering insight, bridging the gap between data analysis and real-world vehicle dynamics understanding.

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1.Introduction

When people think about vehicle performance, they often focus on the engine, aerodynamics, suspension, or braking system. However, every force that allows a vehicle to accelerate, brake, or corner must ultimately pass through four small contact patches where the tires meet the road. No matter how powerful or advanced a vehicle may be, its performance is always limited by the capability of its tires.

For this reason, tires are often regarded as one of the most important components in vehicle dynamics. The way a tire responds to steering inputs, changes in load, or variations in camber can significantly influence the behaviour of a vehicle. A small change in tire performance can affect cornering speed, braking stability, steering feel, and ultimately lap time. Understanding how tires generate force is therefore essential for anyone interested in race car performance and vehicle dynamics.

Unlike many engineering systems that behave in a relatively predictable manner, tires are highly nonlinear. The amount of force a tire can generate depends on several interacting factors, **including vertical load, slip angle, camber angle, temperature, pressure, and road conditions**. As these variables change, tire behaviour changes as well. This complexity is one of the reasons why tire analysis remains one of the most challenging and fascinating areas of vehicle dynamics engineering.

Modern tire testing programs generate vast amounts of experimental data that capture tire behaviour under carefully controlled conditions. While this data contains valuable information, it is often difficult to interpret in its raw form. Thousands of measurements may be recorded during a single test, making it necessary to process, filter, and analyze the data before meaningful conclusions can be drawn. The real engineering challenge lies not in obtaining the data, but in understanding what the data is trying to tell us.

The motivation behind this project was to explore that process firsthand. Rather than treating tire behaviour as a collection of equations or theoretical concepts, this study approaches the subject through experimental data analysis. The objective was to investigate how a racing tire behaves under different operating conditions and to convert raw measurements into meaningful engineering insights.

Throughout this project, various aspects of tire behaviour were examined, including **lateral force generation, load sensitivity, camber sensitivity, cornering stiffness, and self-aligning moment** characteristics. Beyond simply extracting numerical values, the focus was placed on understanding the physical meaning behind the results and relating them to real-world vehicle behaviour.

Ultimately, this project represents a journey from raw tire test data to a deeper understanding of racing tire performance. By combining data analysis with engineering interpretation, the study aims to bridge the gap between measured information and practical vehicle dynamics knowledge while establishing a foundation for future **tire modelling** and **simulation** work.

2. Project Objectives

The primary objective of this project was to develop a practical understanding of racing tire behaviour through the analysis and interpretation of experimental tire test data. Rather than focusing solely on theoretical concepts, the study aimed to investigate how tires generate forces under different operating conditions and how these forces influence overall vehicle performance.

To achieve this objective, a structured data analysis workflow was developed to process raw tire test data and convert it into meaningful engineering information. Through this process, the project sought to identify key tire performance characteristics and understand the physical mechanisms responsible for the observed behaviour.

The specific objectives of this study were:

- To understand the structure and contents of experimental tire test data.
- To investigate the influence of **slip angle**, **vertical load**, and **camber angle** on tire behaviour.
- To identify and characterize the lateral force generation capability of the tire.
- To evaluate **tire load sensitivity** and its implications on vehicle performance.
- To investigate the influence of camber angle on tire force generation.
- To determine **cornering stiffness** and study tire responsiveness to steering inputs.
- To analyze **self-aligning moment** behaviour and its relationship with steering feedback.
- To extract meaningful tire performance parameters for future tire model development.
- To develop a stronger understanding of the relationship between experimental tire data and real-world vehicle dynamics.

Ultimately, the project aims to bridge the gap between raw experimental data and practical engineering understanding, providing a foundation for future tire modelling, vehicle dynamics analysis, and performance simulation work.

3. Tire Data Description

3.1 Overview

The data used in this study was obtained from the **Tire Test Consortium (TTC)**, one of the most **widely used sources** of experimental tire data in vehicle dynamics research and motorsport engineering. The TTC program provides controlled tire testing data recorded under a wide range of operating conditions, including variations in vertical load, slip angle, camber angle, and tire pressure.

The dataset analyzed in this project consists of experimental measurements from a **Hoosier racing tire** and contains the information required to investigate tire force generation and handling characteristics. The availability of multiple operating conditions makes the dataset suitable for studying fundamental tire behaviours such as load sensitivity, camber sensitivity, cornering stiffness, and self-aligning moment generation.

3.2 Test Tire Information

Parameter	Value
Manufacturer	Hoosier Racing Tire
Tire Model	Hoosier 18.0 x 6.0 - 10 R25B
Tire Type	Racing Slick
Data Source	Tire Test Consortium (TTC)
Dataset Used	Run 27
Primary Load Band	600 - 700 N

Table 3.1: Summary of Tire Specifications and Test Dataset

3.3 Sample Raw Data

Before beginning the analysis, the **imported dataset** was inspected to verify the successful data import process and understand the structure of the recorded measurements. Figure 3.1 shows a sample of the raw tire test data obtained from the Tire Test Consortium (TTC) dataset.

ET	V	SA	IA	RL	FX	FY	FZ	MX	MY	MZ	P	N	RE	TSTI	TSTC	TSTO
267.58	40.32	0.04	-0.027	21.381	-71.6	-77.23	-1110.4	0.96	11.6	8.49	83.94	483.85	22.108	52.45	49.06	51.53
267.59	40.32	0.076	-0.036	21.385	-81.55	-108.42	-1118.5	0.08	13.94	5.97	83.79	483.65	22.117	52.44	49.03	51.51
267.6	40.32	0.117	-0.034	21.382	-83.71	-140.36	-1120.5	-0.68	14.53	5.13	83.49	483.51	22.126	52.34	49.04	51.45
267.61	40.3	0.155	-0.037	21.383	-79.37	-169.43	-1122	0.56	13.78	5.92	83.49	483.44	22.116	52.42	49.03	51.4
267.62	40.3	0.206	-0.031	21.383	-71.55	-197.84	-1116.6	1.94	12.14	7.56	83.49	483.71	22.104	52.37	49.01	51.36

Figure 3.1: Sample of Raw Tire Test Data Imported from the TTC Dataset

Figure 3.1 presents a **small portion** of the **imported dataset** containing measurements recorded during tire testing. **Each row** corresponds to a **single measurement point**, while **each column** represents a **specific operating parameter** or tire response. Although only a few rows are shown for illustration purposes, the complete dataset contains thousands of measurements recorded under a wide range of operating conditions. Initial inspection confirmed that all required channels were successfully imported and available for further analysis.

3.4 Available Data Channels

Channel	Full Form	Description
ET	Elapsed Time	Time elapsed during the test
V	Vehicle Speed	Speed of the test wheel/tire
SA	Slip Angle	Angle between wheel direction and actual travel direction
IA	Inclination Angle	Tire camber angle
RL	Loaded Radius	Effective tire radius under load
FX	Longitudinal Force	Force generated in the direction of travel
FY	Lateral Force	Force generated perpendicular to the direction of travel
FZ	Vertical Load	Normal load acting on the tire
MX	Overturning Moment	Moment tending to roll the tire about the longitudinal axis
MY	Rolling Resistance Moment	Moment resisting tire rolling motion
MZ	Self-Aligning Moment	Aligning torque generated by the tire
P	Tire Pressure	Inflation pressure of the tire
N	Wheel Rotational Speed	Rotational speed of the tire/wheel
RE	Effective Radius	Effective rolling radius of the tire
TSTI	Tire Surface Temperature – Inner	Temperature measured at the inner tread region
TSTC	Tire Surface Temperature – Center	Temperature measured at the center tread region
TSTO	Tire Surface Temperature – Outer	Temperature measured at the outer tread region

Table 3.2: Description of Key TTC Data Channels

Among the available channels, particular emphasis was placed on **Slip Angle (SA)**, **Lateral Force (FY)**, **Vertical Load (FZ)**, **Inclination Angle (IA)**, and **Self-Aligning Moment (MZ)**, as these parameters form the basis of tire performance characterization and vehicle dynamics analysis.

4. Initial Dataset Investigation

4.1 Slip Angle Investigation

Slip angle is one of the most important parameters in tire dynamics as it directly influences the generation of lateral force during cornering. **It is defined as the angle between the direction in which the wheel is pointing and the actual direction of travel of the tire.** As slip angle increases, the tire generates increasing lateral force until it reaches its peak grip level, after which the force begins to saturate.

As an initial step in understanding the dataset, the slip angle range was investigated to determine the operating conditions covered during testing. This analysis was important because the available slip angle range determines whether the dataset contains sufficient information to study both the linear and nonlinear regions of tire behaviour.

The investigation showed that the slip angle values within the dataset ranged from **-13.39°** to **+13.16°**.

The dataset contains both positive and negative slip angle values. A positive slip angle occurs when the wheel is pointed in one direction relative to its actual path of travel, while a negative slip angle occurs when the wheel is pointed in the opposite direction. In practical terms, positive and negative slip angles represent tire behaviour during cornering in opposite directions.

Testing across both positive and negative slip angles allows the symmetry of tire behaviour to be evaluated and provides a complete characterization of lateral force generation. For an ideal tire, the magnitude of the generated lateral force is expected to be similar for equal positive and negative slip angles, with the direction of the force simply reversing.

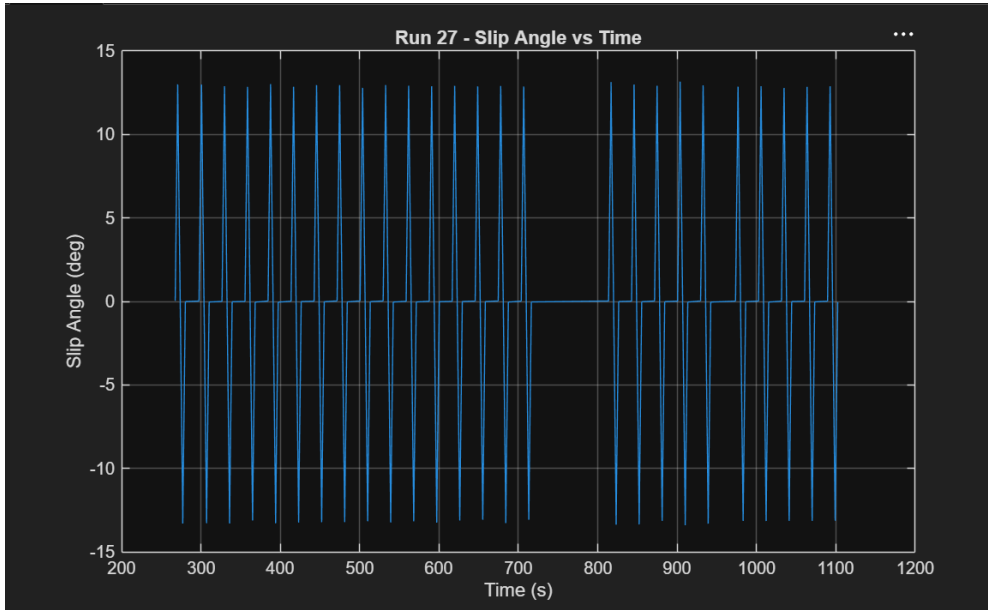


Figure 4.1: Variation of Slip Angle During Tire Testing

An interesting feature of Figure 4.1 is the presence of **short flat regions around 0°** slip angle between successive sweeps. These regions indicate periods where the tire was returned to a **neutral steering** condition before the next test sequence began. Such stabilization periods are commonly used during tire testing to ensure repeatable operating conditions and to separate individual slip-angle sweeps.

The repeated pattern of sweep–stabilize–sweep observed throughout the dataset further confirms that the tire was tested under a controlled and systematic experimental procedure rather than during continuous vehicle operation.

4.2 Camber Angle Investigation

Camber angle, represented by the inclination angle (IA) channel in the dataset, is one of the most influential setup parameters affecting tire performance. **Camber changes the orientation of the tire relative to the road surface and directly influences the size and distribution of the contact patch during cornering.** As a result, variations in camber can significantly affect lateral force generation and overall handling performance.

To understand the operating conditions covered by the dataset, the camber values present during testing were investigated. This analysis was performed to determine whether sufficient data was available to study the influence of camber on tire behaviour and force generation.

Investigation of the dataset revealed that the tire was tested at multiple camber settings, including approximately **0°, 2°, and 4°**. The presence of multiple camber conditions indicates that the test program was designed to evaluate how tire performance changes with wheel inclination.

These camber variations provide the necessary data to investigate camber sensitivity and determine how lateral force generation is influenced by changes in tire inclination.

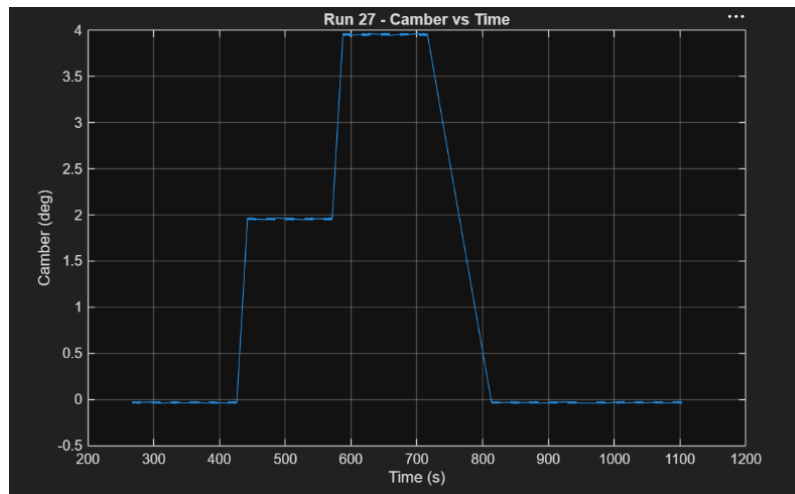


Figure 4.2: Inclination Angle (IA) History of the Test Dataset

4.3 Vertical Load Distribution

Vertical load is one of the most influential parameters affecting tire performance. **The amount of load acting on a tire directly impacts the forces that it can generate and plays a significant role in determining vehicle handling characteristics.** Understanding the distribution of vertical loads present within the dataset is therefore an important step in the tire characterization process.

To investigate the operating conditions covered during testing, the vertical load (FZ) data was analyzed using a **histogram**. This allowed the overall load distribution to be visualized and helped identify the primary load levels at which the tire was tested.

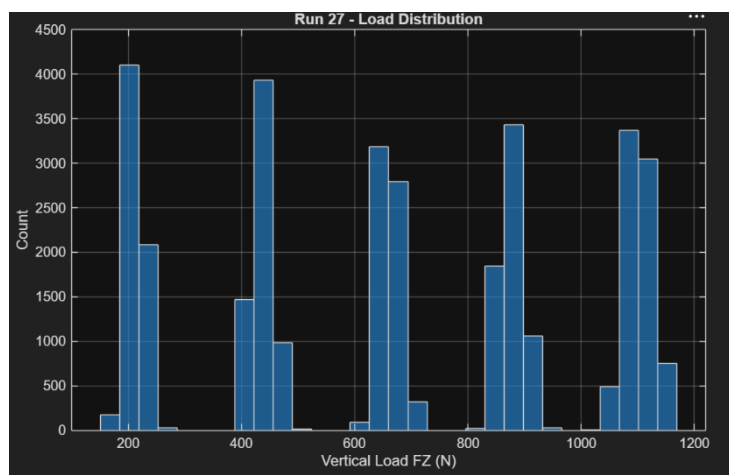


Figure 4.3: Histogram of Vertical Load (FZ) Measurements

Figure 4.3 shows the distribution of vertical load measurements recorded throughout the test. Rather than being spread continuously across the load range, the data is concentrated around

several distinct load levels. This indicates that the tire was tested under controlled loading conditions, allowing its behaviour to be evaluated systematically at specific operating points.

Five major load bands were identified at approximately **200 N**, **450 N**, **650 N**, **900 N**, and **1100 N**. The presence of these distinct load levels is particularly useful because it enables tire behaviour to be compared across a wide range of operating conditions and provides a strong foundation for investigating load sensitivity in later stages of the project.

To determine a representative operating condition for the initial stages of analysis, the overall load distribution was examined further. Statistical analysis of the dataset showed a **mean** vertical load of **674.6 N** and a **median** vertical load of **664.2 N**. Both values are very close to the 650 N load band, indicating that this load level represents the central operating condition of the dataset.

Parameter	Value
Mean Vertical Load	674.6 N
Median Vertical Load	664.2 N
Representative Load Band	650 N

Table 4.1: Statistical Summary of Vertical Load Distribution

The number of data points available within each load band was also evaluated to understand how the test data was distributed across the operating range.

Load Band	Number of Data Points
200 N	6351
450 N	6330
650 N	6260
900 N	5807
1100 N	7412

Table 4.2: Number of Data Points within Each Load Band

An interesting observation is that all five load bands contain a comparable number of data points. This suggests that the tire was tested systematically across the entire load range rather than concentrating heavily on a single operating condition. As a result, each load band contains sufficient data for meaningful analysis.

Although all load levels were suitable for investigation, the **650 N load band** was selected as the primary operating condition for the initial stages of tire characterization. For the **Formula Student vehicle** considered in this project, the estimated static wheel load is approximately **638 N per tire**. Since this value is very close to 650 N, the selected load band provides a realistic representation of the loads expected during normal vehicle operation.

By beginning the analysis at this load level, the results obtained from the tire data can be related more directly to the expected behaviour of the vehicle while still allowing comparisons with the other load bands later in the study.

5. Tire Force Characterization

5.1 Lateral Force Characteristics at the 650 N Load Band

Having identified the operating conditions present within the dataset, the next stage of the investigation focused on examining the lateral force generating capability of the tire. Since lateral force is responsible for producing the cornering forces required for vehicle motion, understanding its relationship with slip angle is fundamental to tire characterization and vehicle dynamics analysis.

For the initial stages of this study, the **650 N load band** was selected as the primary operating condition. This load level was chosen because it closely represents the expected static wheel load of a Formula Student vehicle while also containing a substantial amount of high-quality test data.

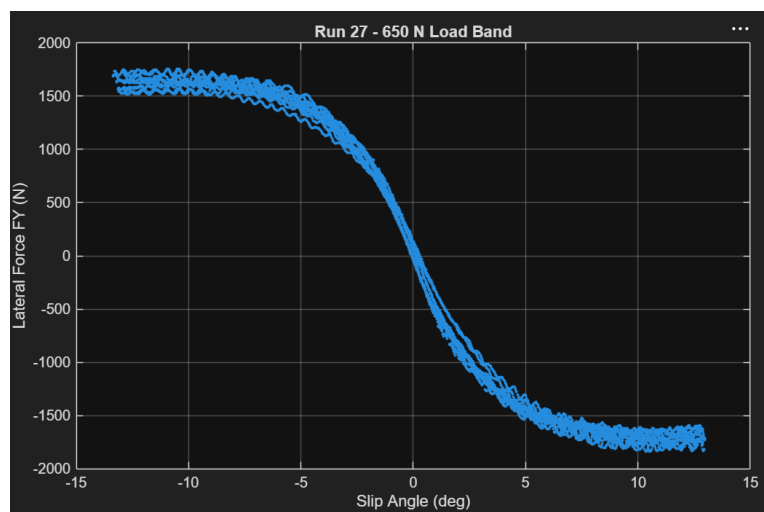


Figure 5.1: Tire Lateral Force Characteristics for the 650 N Load Band

Figure 5.1 illustrates the relationship between slip angle and lateral force for the selected 650 N load band. The curve exhibits the characteristic behaviour expected from a high-performance racing tire and provides valuable insight into the tire's force-generating capability during cornering.

A total of **6260** measurements were available **within** the selected **600–700 N** load range. These measurements were obtained by filtering the dataset based on vertical load, retaining only those data points where the tire load fell within the specified range. The large number of available measurements ensured that the resulting force-slip relationship was well-defined and representative of the tire's behaviour under the selected operating condition.

The smooth and repeatable nature of the curve indicates excellent data quality and confirms that the dataset successfully captures the complete lateral force generation behaviour of the tire. Unlike limited slip-angle tests, the available data extends across the entire operating range of the tire, including the linear, nonlinear, and saturation regions.

Inspection of the curve shows that the tire response can be divided into several distinct operating regions. At small slip angles, the tire operates within a near-linear region where lateral force increases approximately in proportion to slip angle. As slip angle increases further, the tire enters a transitional nonlinear region where force generation continues to increase but at a progressively lower rate. Beyond this region, the tire approaches saturation and eventually reaches its maximum force-generating capability.

The approximate operating regions identified from the force-slip curve are summarized in **Table 5.1**.

Region	Approximate Slip Angle Range	Characteristics
Linear Region	0° to 2°	Lateral force increases approximately proportionally with slip angle
Transitional Region	2° to 8°	Force continues to increase, but the relationship becomes nonlinear
Saturation Region	8° to 13°	Tire approaches peak grip and additional slip angle produces limited force increase
Peak Force Region	~10° to 11°	Maximum lateral force generation occurs

Table 5.1: Approximate Tire Operating Regions at the 650 N Load Band

The existence of these regions highlights the inherently nonlinear nature of tire behaviour. From a vehicle dynamics perspective, the linear region is particularly important because it forms the basis for cornering stiffness calculations, while the saturation region defines the ultimate grip capability of the tire. Understanding the transition between these regions is essential for predicting vehicle handling behaviour and developing accurate tire models.

The presence of all major operating regions within a single dataset confirms that the selected load band is suitable for detailed tire characterization and provides a strong foundation for extracting peak force characteristics and other key tire performance parameters.

5.2 Peak Lateral Force Extraction

One of the primary objectives of tire characterization is to determine the maximum force that a tire can generate under a given operating condition. This peak force represents the highest level of cornering performance available from the tire and provides valuable insight into its overall grip capability.

To identify the peak performance point, the filtered 650 N load band dataset was examined and the maximum lateral force value was extracted. The analysis showed that the tire generated a peak lateral force of **1840.2 N** at a slip angle of **10.37°** under a vertical load of **681.8 N**.

The corresponding friction coefficient was calculated using the ratio of lateral force to vertical load and was found to be **2.699**. This indicates that the tire was capable of generating a lateral force approximately **2.7** times greater than the applied wheel load under the tested conditions, highlighting the high grip levels achievable by a modern racing tire.

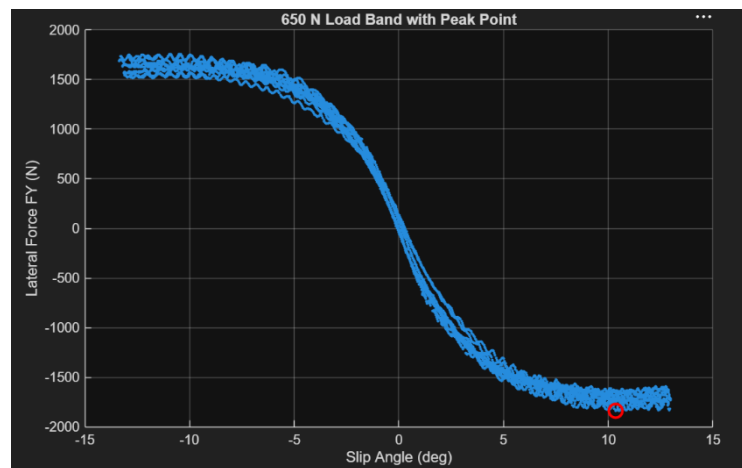


Figure 5.2: Peak Lateral Force Point on the Tire Force–Slip Angle Curve

Figure 5.2 highlights the operating condition at which the maximum lateral force was generated within the selected 650 N load band. The highlighted point corresponds to the highest measured lateral force obtained from the filtered dataset and represents the peak performance capability of the tire under the tested conditions.

The key parameters associated with the peak force condition are summarized in **Table 5.2**

Parameter	Value
Peak Lateral Force (FY)	1840.2 N
Peak Slip Angle (SA)	10.37°
Vertical Load (FZ)	681.8 N
Peak Friction Coefficient (μ)	2.699

Table 5.2: Peak Tire Performance Parameters for the 650 N Load Band

6. Load Sensitivity Analysis

6.1 Introduction

One of the most important characteristics of a racing tire is its sensitivity to changes in vertical load. **While increasing the load acting on a tire generally increases the amount of force it can generate, this increase is not proportional.** As a result, the tire becomes **less efficient** at higher loads, a phenomenon commonly referred to as tire load sensitivity.

Understanding load sensitivity is particularly important in vehicle dynamics because it directly influences the effects of weight transfer during cornering, braking, and acceleration. If tire force increased proportionally with load, weight transfer would have little impact on overall vehicle performance. However, because tire force increases at a diminishing rate, excessive load transfer can reduce the total grip available to the vehicle.

To investigate this behaviour, peak lateral force and friction coefficient values were extracted from each of the identified load bands. By comparing tire performance across multiple operating loads, it becomes possible to evaluate how efficiently the tire converts vertical load into usable lateral force.

6.2 Peak Tire Performance Across Load Bands

Load Band (N)	Peak FY (N)	Peak μ	Peak SA (deg)
200	682.1	2.744	12.53
450	1272.1	2.745	12.08
650	1840.2	2.699	10.37
900	2341.3	2.557	10.65
1100	2857.1	2.499	10.66

Table 6.1: Peak Tire Performance Parameters Across Load Bands

The peak performance parameters extracted from each load band are summarized in **Table 6.1**. As vertical load increased, the tire generated progressively larger lateral forces, with peak lateral force increasing from **682.1 N** at the **200 N** load band to **2857.1 N** at the **1100 N** load band.

Although the increase in peak force is substantial, the friction coefficient does not follow the same trend. Instead, peak μ gradually decreases as load increases. This observation suggests that while the tire becomes capable of generating greater absolute force at higher loads, it becomes less efficient at converting vertical load into usable lateral force.

6.3 Effect of Vertical Load on Peak Lateral Force

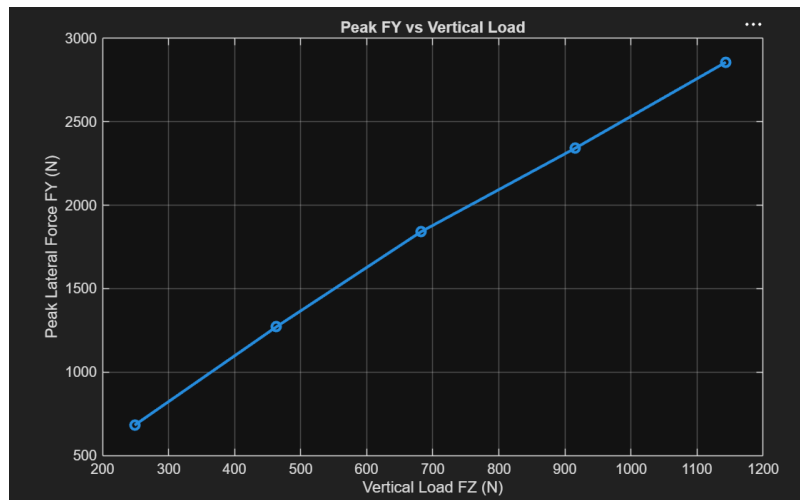


Figure 6.1: Variation of Peak Lateral Force with Vertical Load

Figure 6.1 shows the variation of peak lateral force with vertical load for the tested tire.

A clear positive relationship can be observed, with peak lateral force increasing consistently as the applied vertical load increases.

At the lowest load condition of approximately **200 N**, the tire generated a peak lateral force of **682.1 N**. As the vertical load increased through the successive load bands, the peak force also increased, reaching 1272.1 N at 450 N, 1840.2 N at 650 N, 2341.3 N at 900 N, and 2857.1 N at the highest load condition of 1100 N.

This trend indicates that the tire is capable of generating greater absolute cornering force as additional load is applied. The increase in force occurs because a larger vertical load increases the interaction between the tire and the road surface, allowing greater forces to be developed within the contact patch.

Although the relationship appears nearly linear over the tested load range, closer examination shows that the increase in force is not perfectly proportional to the increase in load. For example, increasing the vertical load from approximately 200 N to 1100 N results in a more than fourfold increase in lateral force, but the rate of force generation gradually decreases at higher loads. This behaviour is characteristic of racing tires and highlights the nonlinear nature of tire performance.

From a vehicle dynamics perspective, the results confirm that increasing vertical load generally increases the maximum cornering force available from a tire. However, absolute force generation alone does not fully describe tire performance. To determine whether the tire becomes more or less efficient at generating force under higher loading conditions, the corresponding variation in friction coefficient must also be examined.

The peak force values extracted from each load band are summarized in Table 6.1, while **Figure 6.1** provides a visual representation of the overall trend.

6.4 Effect of Vertical Load on Friction Coefficient

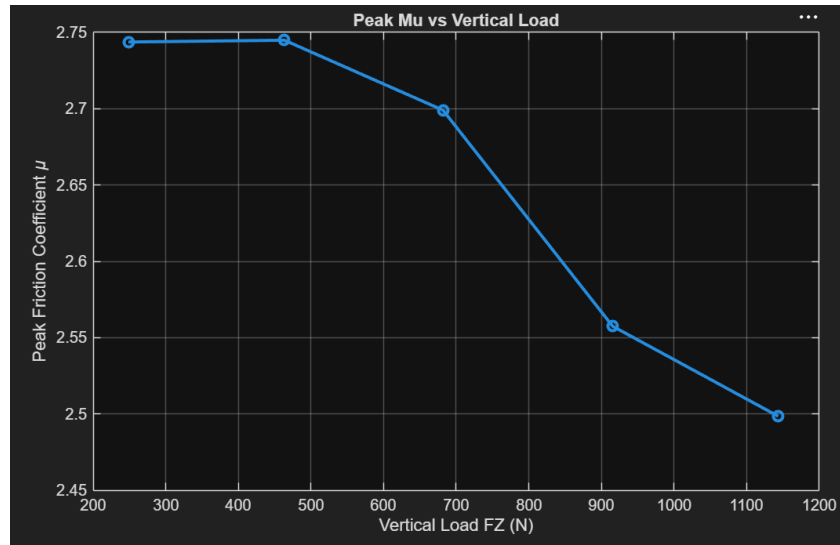


Figure 6.2: Variation of Peak Friction Coefficient with Vertical Load

Figure 6.2 shows the variation of peak friction coefficient with vertical load for the tested tire. Unlike the trend observed in peak lateral force generation, the friction coefficient decreases as vertical load increases.

At the lower load conditions, the tire achieved a peak friction coefficient of approximately **2.74**. As the vertical load increased through the successive load bands, the peak friction coefficient gradually decreased, reaching approximately **2.50** at the highest load condition. This reduction can be observed consistently across the tested load range.

The decreasing trend indicates that although the tire is capable of generating larger absolute lateral forces at higher loads, it becomes less efficient at converting vertical load into usable grip. In other words, the increase in lateral force does not occur at the same rate as the increase in vertical load.

This behaviour is known as tire load sensitivity and is a fundamental characteristic of pneumatic tires. As additional load is applied to the tire, the contact patch and tire deformation increase, but the resulting increase in force generation is not perfectly proportional. Consequently, the friction coefficient decreases as load increases.

The trend observed in **Figure 6.2** complements the results presented in **Figure 6.1**. **While higher vertical loads allow the tire to generate greater cornering forces, they also reduce the tire's overall force generation efficiency.** Therefore, evaluating tire performance requires consideration of both absolute force generation and friction coefficient behaviour.

The results obtained from the TTC dataset clearly demonstrate the presence of tire load sensitivity, with peak μ decreasing from approximately 2.74 at low loads to approximately 2.50 at the highest load condition.

7. Camber Sensitivity Analysis

Following the investigation of tire load sensitivity, the next stage of the analysis focused on evaluating the influence of camber angle on tire performance. Camber angle is an important vehicle setup parameter that affects the orientation of the tire relative to the road surface and can significantly influence the shape and pressure distribution of the tire contact patch during cornering. As a result, changes in camber can alter the tire's ability to generate lateral force and therefore have a direct impact on vehicle handling and cornering performance.

To investigate this behaviour, the 650 N load band was selected as the reference operating condition and the peak lateral force generated at different camber angles was compared. This approach allowed the effect of camber to be isolated while maintaining a consistent vertical load condition throughout the analysis.

7.1 Data Availability Across Camber Conditions

Before evaluating the effect of camber angle on tire performance, the number of available data samples at each camber condition was examined. This step was necessary to ensure that sufficient data existed at each operating condition and that meaningful comparisons could be made between the different camber settings.

The data distribution within the selected 650 N load band is summarized in Table 7.1.

Camber Angle	Number of Data Points
0°	3740
2°	1266
4°	1254

Table 7.1: Data Distribution Across Camber Conditions

Although the number of measurements varied between the three camber conditions, all settings contained a substantial amount of data. This provided confidence that the extracted performance parameters were representative of the tire's behaviour and suitable for comparative analysis.

7.2 Effect of Camber on Peak Lateral Force

The influence of camber angle on tire performance was evaluated by comparing the peak lateral force generated at each camber condition within the selected load band. The results are presented in **Figure 7.1**.

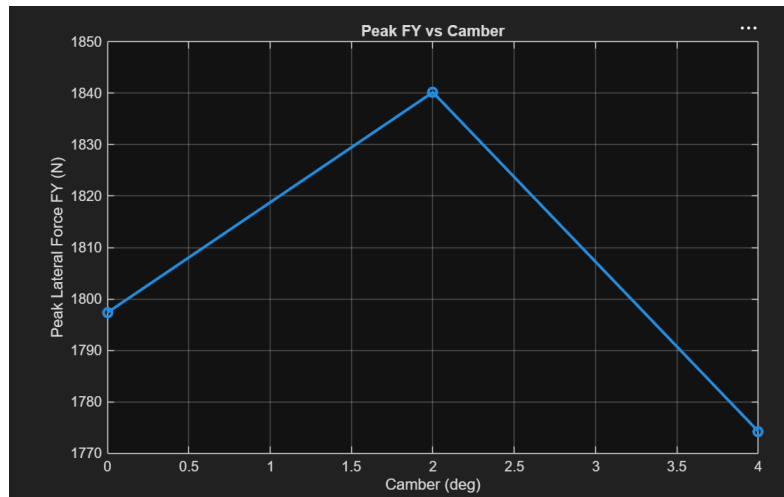


Figure 7.1: Variation of Peak Lateral Force with Camber Angle

Figure 7.1 shows that tire performance varied with camber angle and that the maximum lateral force did not occur at either extreme of the tested camber range. Instead, the highest measured force was achieved at an intermediate camber condition.

Peak lateral force increased from 1797.3 N at 0° camber to 1840.2 N at 2° camber, representing the highest force generated within the dataset. As camber was increased further to 4°, peak lateral force decreased to 1774.2 N.

Camber Angle	Peak Lateral Force (FY)
0°	1797.3 N
2°	1840.2 N
4°	1774.2 N

Table 7.2: Peak Lateral Force at Different Camber Conditions

The results indicate that a moderate amount of camber improved lateral force generation under the tested conditions. Increasing camber from 0° to 2° produced a measurable increase in peak force, suggesting improved utilization of the tire contact patch during cornering. However, further increases in camber resulted in a reduction in force generation, indicating that excessive camber may reduce the effective contact area available for producing lateral force.

Although the differences between the three operating conditions are relatively small, such variations can still be important in motorsport applications where small improvements in tire performance can contribute to measurable gains in vehicle handling and lap time.

7.3 Optimum Camber Condition

The camber sensitivity study identified approximately 2° camber as the optimum operating condition within the tested range. The maximum measured lateral force of 1840.2 N occurred at a camber angle of 1.96°, corresponding to a slip angle of 10.37° and a vertical load of 681.8 N.

The operating conditions associated with the peak performance point are summarized in Table 7.3.

Parameter	Value
Vertical Load	681.8 N
Slip Angle	10.37°
Camber Angle	1.96°
Peak Lateral Force	1840.2 N
Peak Friction Coefficient	2.699

Table 7.3: Operating Conditions at Peak Tire Performance

Based on the available data, the combination of approximately 682 N vertical load, 10.37° slip angle, and 2° camber produced the highest measured tire performance within the Run 27 dataset. Under these conditions, the tire generated a peak lateral force of 1840.2 N and a corresponding peak friction coefficient of 2.699. Consequently, this operating condition was considered the optimum test condition for peak force extraction within the selected load band.

It should be noted that the calculated friction coefficient represents the maximum measured force-to-load ratio obtained from the TTC dataset under controlled laboratory testing conditions. While the value is relatively high compared to the friction coefficients commonly assumed in simplified vehicle dynamics models, it represents a peak experimental result rather than a continuously achievable operating condition. Therefore, the extracted value should be interpreted as an indicator of the tire's maximum measured performance rather than its average on-track behaviour.

The results demonstrate that camber angle plays an important role in tire performance and that an appropriate amount of camber can improve lateral force generation. Increasing camber from 0° to approximately 2° produced the highest measured grip, while further increases in camber resulted in a reduction in peak lateral force. This indicates that there exists an optimum camber range within which the tire can utilize the contact patch most effectively.

These findings highlight the importance of carefully selecting camber settings when optimizing vehicle performance and provide valuable insight for future tire modelling, vehicle setup decisions, and lap time simulation development.

8. Cornering Stiffness Analysis

While peak lateral force describes the maximum grip capability of a tire, it does not indicate how rapidly that grip is generated as steering input is applied. To evaluate the tire's responsiveness, a cornering stiffness analysis was performed.

Cornering stiffness is defined as the slope of the lateral force versus slip angle curve in the approximately linear operating region near zero slip angle. It represents the rate at which lateral force is generated for a given change in slip angle and is one of the most important parameters used in vehicle dynamics modelling.

A tire with a high cornering stiffness generates lateral force rapidly for small steering inputs and therefore responds more quickly to driver commands. Conversely, a tire with a lower cornering stiffness requires larger slip angles to generate the same force and may feel less responsive during transient manoeuvres.

To determine cornering stiffness, the lateral force behaviour near zero slip angle was isolated and examined within the selected 650 N load band.

8.1 Initial Linear Region Investigation ($\pm 2^\circ$)

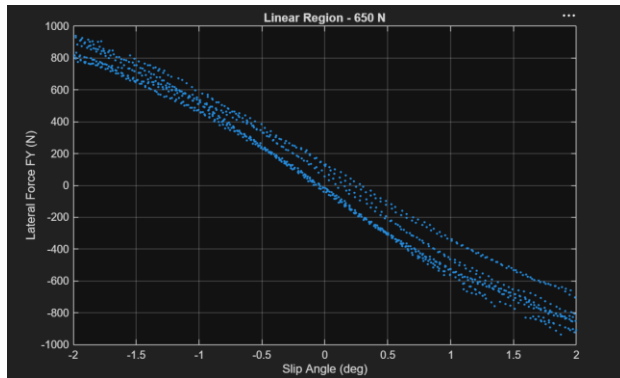


Figure 8.1: Lateral Force Behaviour within $\pm 2^\circ$ Slip Angle

Figure 8.1 shows the relationship between lateral force and slip angle within the $\pm 2^\circ$ region for the selected 650 N load band. This region was isolated to investigate the tire's behaviour near zero slip angle, where cornering stiffness is typically evaluated.

The plot exhibits an approximately linear relationship between slip angle and lateral force, indicating that the tire generates force nearly proportionally to steering input within this operating range. However, slight curvature can be observed near the boundaries of the selected region, particularly as the slip angle approaches $\pm 2^\circ$. This suggests that the tire is beginning to transition away from purely linear behaviour and enter the early stages of the nonlinear region.

Although the overall response remains largely linear, the presence of these small nonlinear effects indicates that the $\pm 2^\circ$ range may not represent the tire's true initial response with maximum accuracy. Consequently, a narrower slip angle window was investigated to isolate

the most linear portion of the tire response and obtain a more representative cornering stiffness value.

The observations from Figure 8.1 confirmed that the tire exhibits strong sensitivity to steering input near the origin and provided the basis for the more detailed analysis performed within the $\pm 1^\circ$ slip angle region.

8.2 Identification of the Linear Operating Region

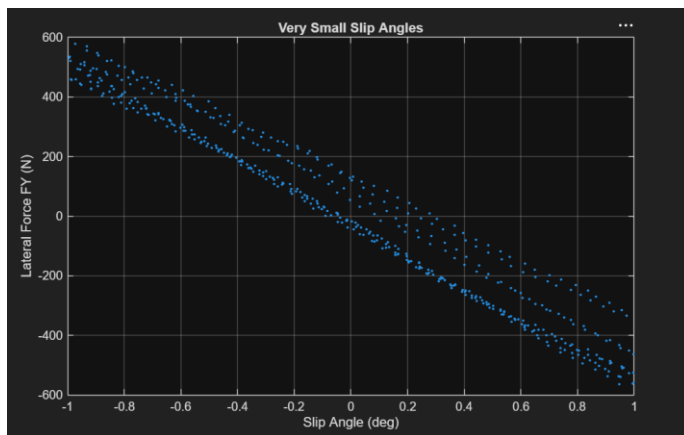


Figure 8.2: Lateral Force Behaviour within $\pm 1^\circ$ Slip Angle

Figure 8.2 shows the lateral force versus slip angle relationship within the reduced $\pm 1^\circ$ slip angle range for the selected 650 N load band. Compared to the broader $\pm 2^\circ$ investigation, the tire response within this narrower region appears significantly more linear, with the majority of the data following a nearly straight-line trend.

The reduction in slip angle range was performed to isolate the tire's initial response to steering input and minimize the influence of nonlinear effects. Visual inspection of the figure shows that the slight curvature observed near the boundaries of the $\pm 2^\circ$ region is largely absent within the $\pm 1^\circ$ window. Although a small amount of scatter remains due to measurement noise and variations in test conditions, the overall behaviour can be considered approximately linear.

This observation confirms that the tire generates lateral force almost proportionally with slip angle near the origin. Since cornering stiffness is defined as the initial slope of the lateral force versus slip angle curve, the identified $\pm 1^\circ$ region provides a more representative operating range for its calculation.

The highly linear nature of the response indicates that the tire is operating well below its peak force condition and has not yet entered the nonlinear region associated with force saturation. Consequently, the $\pm 1^\circ$ slip angle range was selected as the preferred region for cornering stiffness evaluation.

8.3 Cornering Stiffness Calculation

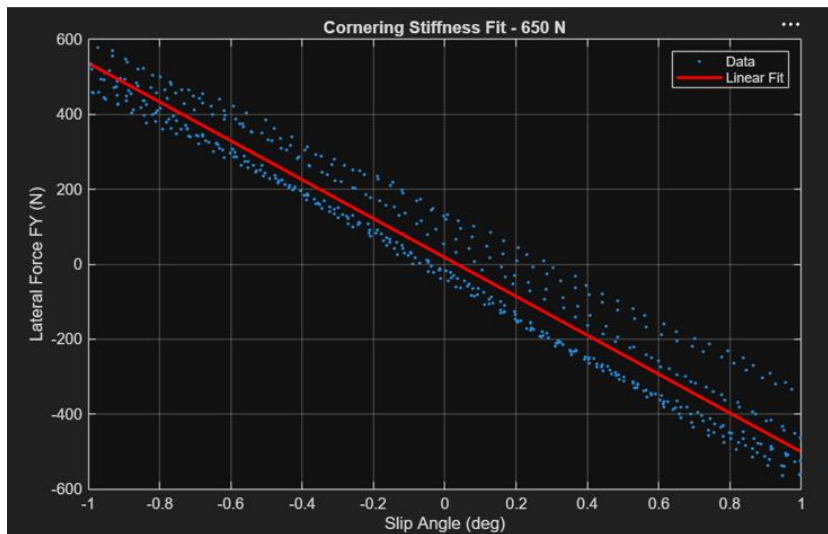


Figure 8.3: Linear Regression Used for Cornering Stiffness Calculation

Cornering stiffness was calculated using the identified linear operating region within the selected 650 N load band. A straight-line fit was applied to the lateral force versus slip angle data within the $\pm 1^\circ$ range, allowing the initial slope of the tire response to be determined.

Cornering stiffness is defined as the rate of change of lateral force with respect to slip angle and represents the tire's responsiveness to steering input. A larger cornering stiffness value indicates that the tire is capable of generating lateral force more rapidly for a given change in slip angle.

Figure 8.3 shows the linear regression applied to the selected data points. The close agreement between the fitted line and the measured data confirms that the $\pm 1^\circ$ slip angle range provides a suitable approximation of the tire's linear operating region.

The calculated cornering stiffness value is summarized in Table 8.1.

Parameter	Value
Load Band	600–700 N
Slip Angle Range	$\pm 1^\circ$
Analysis Method	Linear Regression
Cornering Stiffness	518.8 N/deg

Table 8.1: Cornering Stiffness Calculation Results

The analysis produced a cornering stiffness value of 518.8 N/deg. This indicates that, within the linear operating region, the tire generates approximately 519 N of lateral force for every one-degree increase in slip angle.

From a vehicle dynamics perspective, cornering stiffness describes the tire's initial steering response and plays a major role in determining vehicle handling characteristics. While peak lateral force defines the ultimate grip capability of the tire, cornering stiffness describes how quickly that grip is developed as steering input is applied. Consequently, both parameters are required to fully characterize tire performance.

8.4 Effect of Vertical Load on Cornering Stiffness

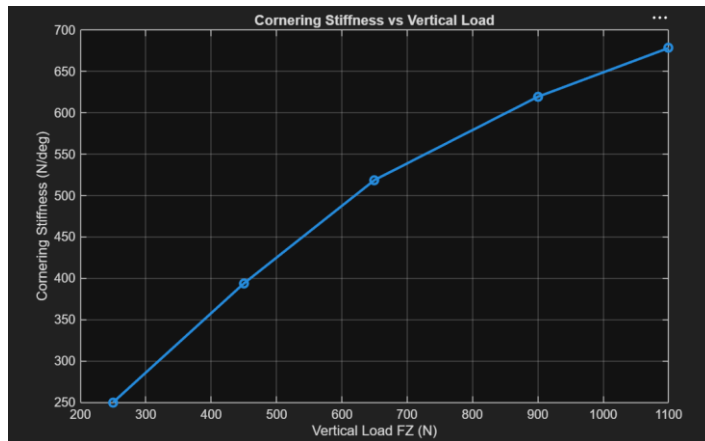


Figure 8.4: Variation of Cornering Stiffness with Vertical Load

Figure 8.4 illustrates the variation of cornering stiffness with vertical load for the tested tire. A clear positive relationship can be observed, with cornering stiffness increasing consistently as the applied load increases.

The results show that cornering stiffness increased from approximately 250.1 N/deg at the 250 N load condition to 678.4 N/deg at the 1100 N load condition. This indicates that the tire generates lateral force more rapidly for a given change in slip angle when operating under higher vertical loads.

The increase in cornering stiffness suggests that the tire becomes more responsive as load increases. Under higher loading conditions, the tire develops larger forces within the contact patch and therefore reacts more aggressively to steering inputs. As a result, a heavily loaded tire requires a smaller change in slip angle to generate a given amount of lateral force compared to a lightly loaded tire.

Although cornering stiffness continues to increase throughout the tested load range, the rate of increase gradually decreases at higher loads. The improvement observed between the lower load bands is significant, whereas the increase between 900 N and 1100 N is comparatively smaller. This indicates that the benefits of additional load begin to diminish as the tire approaches higher operating loads.

The cornering stiffness values extracted from each load band are summarized in Table 8.2.

Load Band (N)	Cornering Stiffness (N/deg)
250	250.1
450	393.8
650	518.8
900	619.5
1100	678.4

Table 8.2: Cornering Stiffness Across Load Bands

The observed trend is consistent with the tire behaviour identified throughout the study and demonstrates that tire responsiveness improves with increasing vertical load. However, these results must be considered together with the load sensitivity analysis presented earlier, where the friction coefficient was shown to decrease as load increased. Therefore, while higher loads improve tire responsiveness, they do not necessarily improve the overall efficiency of force generation.

8.5 Relationship Between Cornering Stiffness and Load Sensitivity

The cornering stiffness analysis and load sensitivity analysis provide two different perspectives on tire performance. While load sensitivity describes how efficiently a tire converts vertical load into lateral force, cornering stiffness describes how rapidly that force is generated in response to steering input.

The load sensitivity analysis showed that increasing vertical load resulted in higher peak lateral force but a lower friction coefficient. In contrast, the cornering stiffness analysis demonstrated that increasing vertical load caused the tire to generate lateral force more rapidly, resulting in higher responsiveness.

These observations reveal an important characteristic of tire behaviour. As the tire becomes more heavily loaded, it responds more quickly to steering inputs and develops lateral force at a faster rate. However, the tire simultaneously becomes less efficient at converting vertical load into usable grip.

This relationship is summarized conceptually in Table 8.3.

Increasing Vertical Load	Effect
Peak Lateral Force	Increases
Cornering Stiffness	Increases
Friction Coefficient (μ)	Decreases
Tire Responsiveness	Increases
Tire Efficiency	Decreases

Table 8.3: Influence of Vertical Load on Tire Characteristics

From a vehicle dynamics perspective, this finding, highlights the trade-off that exists between tire responsiveness and tire efficiency. A heavily loaded tire is capable of generating force more rapidly and therefore responds more aggressively to steering inputs. However, the reduction in friction coefficient means that the tire becomes less effective at producing grip relative to the applied load.

This behaviour helps explain why excessive weight transfer can negatively affect vehicle performance. During cornering, the outside tire experiences a large increase in vertical load and becomes more responsive, but the associated reduction in tire efficiency limits the total grip available at the axle. Consequently, vehicle setup parameters such as suspension stiffness, roll stiffness distribution, track width, and center of gravity height must be carefully optimized to balance responsiveness and grip generation.

The results obtained from the TTC dataset clearly demonstrate that tire performance cannot be evaluated using a single parameter alone. Peak lateral force, friction coefficient, and cornering stiffness each describe different aspects of tire behaviour and must be considered together when analyzing vehicle handling characteristics and developing tire models.

9. Self-Aligning Moment Analysis

9.1 Fundamentals of Self-Aligning Moment

When a tire operates at a non-zero slip angle, the tire does not immediately slide across the road surface. Instead, the rubber within the contact patch deforms as the wheel rolls forward. This deformation creates lateral forces that allow the tire to generate cornering grip. The distribution of these forces within the contact patch is not uniform. **During cornering, a larger portion of the force is generated behind the center of the contact patch.** As a result, a restoring torque is produced that attempts to align the tire with its direction of travel. This restoring torque is known as the self-aligning moment (MZ).

At small slip angles, the contact patch remains largely elastic and the tire generates a strong self-aligning effect. The steering system therefore experiences a noticeable restoring torque, providing the driver with steering weight and feedback. As slip angle increases, the tire continues to build lateral force and the self-aligning moment initially increases. However, as the tire approaches its grip limit, portions of the contact patch begin to slide rather than deform elastically. This reduces the tire's ability to generate restoring torque and causes the self-aligning moment to decrease.

Consequently, the maximum self-aligning moment typically occurs at a smaller slip angle than the maximum lateral force. While the tire may continue generating additional cornering force, the steering feedback available to the driver begins to reduce. This behaviour is one of the most important characteristics of pneumatic tires because it provides the driver with information about the available grip level. As the tire approaches its operating limit, the reduction in self-aligning moment causes the steering wheel to feel lighter, providing an indication that the tire is nearing saturation.

9.2 Self-Aligning Moment Characteristics

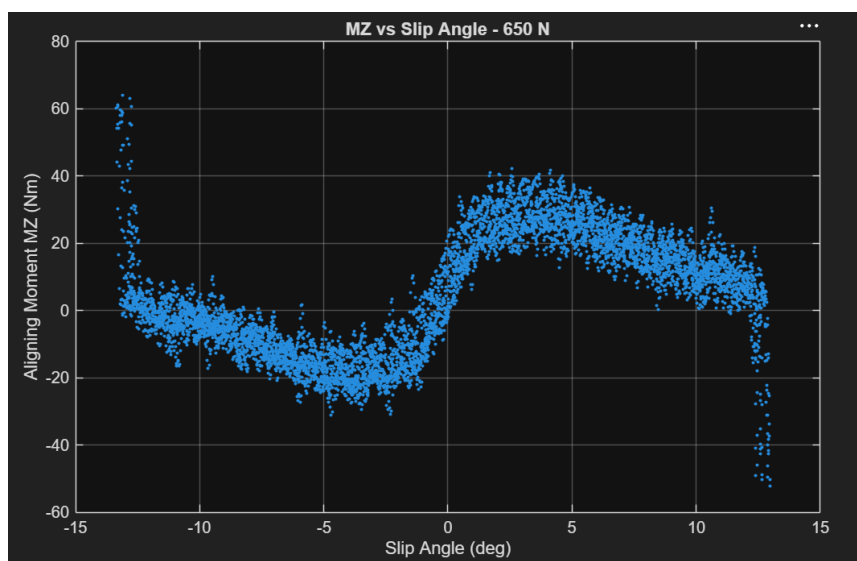


Figure 9.1: Relationship Between Self-Aligning Moment and Slip Angle

Figure 9.1 illustrates the variation of self-aligning moment with slip angle for the selected 650 N load band. Unlike the lateral force curve presented earlier, the self-aligning moment does not continuously increase with slip angle. Instead, it exhibits a distinct peak before gradually decreasing as slip angle increases further.

At small slip angles, the tire generates a relatively large aligning moment due to elastic deformation within the contact patch. As slip angle increases, the aligning moment initially rises, indicating an increase in steering feedback and self-centering behaviour.

The maximum self-aligning moment occurs at moderate slip angles, after which the aligning moment begins to decrease despite the continued increase in lateral force. This reduction occurs because portions of the contact patch begin to slide as the tire approaches its operating limit, reducing the effectiveness of the restoring torque generated within the contact patch.

The overall shape of the curve demonstrates that steering feedback and cornering force do not develop in the same manner. While lateral force continues to build toward its peak value, the self-aligning moment reaches a maximum earlier and subsequently decreases. This behaviour is characteristic of high-performance pneumatic tires and provides valuable information regarding the approach to tire saturation.

9.3 Peak Self-Aligning Moment

The self-aligning moment data was further examined to identify the operating condition corresponding to the maximum steering feedback generated by the tire. Inspection of **Figure 9.1** showed that the peak self-aligning moment occurred at a slip angle of approximately 2.58° .

Parameter	Value
Peak Self-Aligning Moment (MZ)	42.28 Nm
Slip Angle at Peak MZ	2.58°
Slip Angle at Peak FY	10.37°

Table 9.1: Peak Self-Aligning Moment Parameters

An important observation is that the maximum self-aligning moment occurs at a significantly smaller slip angle than the maximum lateral force. While peak self-aligning moment was reached at approximately 2.58° slip angle, peak lateral force occurred much later at approximately 10.37° slip angle.

This behaviour is characteristic of high-performance pneumatic tires and reflects the changes occurring within the tire contact patch as slip angle increases. At relatively small slip angles, the contact patch remains largely within the elastic deformation region, allowing the tire to generate a strong restoring torque and significant steering feedback. As slip angle increases further, portions of the contact patch begin to slide, reducing the tire's self-centering capability and causing the aligning moment to decrease.

Although lateral force continues to increase beyond this point, the tire becomes progressively less effective at generating restoring torque. Consequently, the peak self-aligning moment occurs well before the peak lateral force condition.

The results obtained from the Run 27 dataset indicate that the tire delivers its strongest steering feedback at approximately 2.58° slip angle, whereas maximum grip is achieved closer to 10° slip angle. This difference highlights the fact that steering feel and tire grip are related but do not reach their maximum values simultaneously.

9.4 Driver Feedback Interpretation

The relationship between self-aligning moment and slip angle provides valuable insight into the information available to a driver during cornering. **While lateral force determines the grip available to the vehicle, self-aligning moment influences the steering feedback experienced through the steering wheel.**

As steering input is applied and slip angle begins to increase, the self-aligning moment also increases. This results in a stronger restoring torque acting on the steering system and causes the steering wheel to feel heavier and more connected. During this stage, the tire is operating within a stable region and continues to build both lateral force and steering feedback.

As the tire approaches the slip angle corresponding to peak self-aligning moment, steering feedback reaches its maximum level. Beyond this point, the aligning moment begins to decrease even though lateral force continues to increase. Consequently, the steering wheel may begin to feel lighter despite the tire still generating additional cornering force.

This reduction in steering feedback serves as an important indication that the tire is approaching its operating limit. Experienced drivers often use these changes in steering feel to judge the available grip and determine how close the vehicle is to the limit of adhesion. In many cases, the reduction in steering weight provides an earlier warning than the reduction in lateral force itself.

The results obtained from the Run 27 dataset clearly demonstrate this behaviour. Peak self-aligning moment occurred at approximately 2.58° slip angle, whereas peak lateral force occurred at approximately 10.37° slip angle. Therefore, the maximum steering feedback available to the driver is achieved significantly before the tire reaches its maximum grip capability.

This observation is one of the most important findings of the self-aligning moment analysis, as it directly connects measured tire data to real-world driver perception and vehicle handling behaviour.

10. Conclusions

The objective of this project was to analyze experimental racing tire test data and extract meaningful tire performance characteristics for vehicle dynamics analysis and future tire model development. Through a systematic investigation of slip angle behaviour, vertical load effects, camber sensitivity, cornering stiffness, and self-aligning moment characteristics, the raw test data was transformed into a set of engineering parameters that describe the tire's behaviour under a range of operating conditions.

The analysis began with an investigation of the available operating conditions within the dataset. Slip angle sweeps, camber settings, and vertical load distributions were examined to understand the structure of the test data and identify representative operating conditions. A primary load band of approximately 650 N was selected for detailed tire characterization as it closely represented the expected wheel load of a Formula Student vehicle.

Lateral force analysis showed that the tire exhibited the expected nonlinear behaviour characteristic of a racing tire, consisting of linear, transitional, and saturation regions. Within the selected load band, a peak lateral force of 1840.2 N was generated at a slip angle of 10.37° . The corresponding operating condition occurred at a vertical load of 681.8 N and a camber angle of approximately 2° , producing a peak measured friction coefficient of 2.699 under the controlled TTC test conditions.

Load sensitivity analysis confirmed that increasing vertical load increased the absolute lateral force generated by the tire. However, the friction coefficient decreased with increasing load, demonstrating the nonlinear relationship between vertical load and tire force generation. This behaviour highlighted the importance of load transfer management and reinforced fundamental vehicle dynamics principles regarding tire efficiency.

Camber sensitivity analysis identified approximately 2° camber as the optimum operating condition within the tested range. Moderate camber improved lateral force generation, while additional camber produced diminishing benefits and ultimately reduced peak force capability. This observation demonstrated the importance of camber optimization when configuring a racing vehicle.

Cornering stiffness analysis provided insight into the tire's responsiveness to steering input. A cornering stiffness value of 518.8 N/deg was calculated for the selected load band, and further analysis showed that cornering stiffness increased with vertical load. The results demonstrated that while heavily loaded tires become more responsive, they also experience reduced force generation efficiency, highlighting an important trade-off between responsiveness and grip utilization.

Self-aligning moment analysis revealed that maximum steering feedback occurred at a slip angle of approximately 2.58° , significantly earlier than the slip angle corresponding to peak lateral force. This finding demonstrated that steering feel and maximum grip do not occur simultaneously and provided valuable insight into how drivers perceive the approach to the tire's operating limit.

Overall, the study successfully extracted the key tire characteristics required for the development of an initial tire model. More importantly, it demonstrated how raw experimental tire data can be transformed into meaningful engineering information that can be applied to vehicle dynamics analysis, tire modelling, and future lap time simulation development. The results obtained throughout this project provide a strong foundation for the next stage of work, which will focus on constructing and validating a mathematical tire model based on the extracted characteristics.

11. References

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- [3] Milliken Research Associates, *Data Analysis Reference Manual (Tire Testing Consortium)*, Milliken Research Associates, USA.
- [4] Society of Automotive Engineers (SAE), *SAE J670 – Vehicle Dynamics Terminology*, SAE International, Warrendale, Pennsylvania, USA.
- [5] Gillespie, T. D., *Fundamentals of Vehicle Dynamics*, SAE International, Warrendale, Pennsylvania, USA, 1992.
- [6] Hoosier Racing Tire Corp., *Hoosier Racing Tire Technical Information and Tire Data Resources*.
- [7] Tire Testing Consortium (TTC), *Run 27 Experimental Tire Test Data*, Calspan Tire Research Facility.

Appendix A – MATLAB Scripts

This appendix contains the MATLAB scripts developed and used throughout the tire characterization process. The scripts were used for data filtering, tire force analysis, load sensitivity investigations, cornering stiffness calculations, and self-aligning moment extraction from the TTC Run 27 dataset.

```
clc
clear
close all

%% Import Data

filename = 'Tiredatarun27 (1).dat';

data = readtable(filename);

data(1:5,:)
%% Variable Names

data.Properties.VariableNames
%% Basic Statistics

fprintf('\n===== BASIC STATISTICS =====\n\n')

fprintf('Minimum SA = %.2f deg\n',min(data.SA))
fprintf('Maximum SA = %.2f deg\n',max(data.SA))

fprintf('Minimum IA = %.2f deg\n',min(data.IA))
fprintf('Maximum IA = %.2f deg\n',max(data.IA))

fprintf('Minimum FZ = %.0f N\n',min(abs(data.FZ)))
fprintf('Maximum FZ = %.0f N\n',max(abs(data.FZ)))
%% Slip Angle vs Time

figure

plot(data.ET,data.SA)

xlabel('Time (s)')
ylabel('Slip Angle (deg)')

title('Run 27 - Slip Angle vs Time')
%% Camber vs Time

figure

plot(data.ET,data.IA)

xlabel('Time (s)')
ylabel('Camber (deg)')

title('Run 27 - Camber vs Time')

grid on
%% Load Distribution

figure

histogram(abs(data.FZ),30)

xlabel('Vertical Load FZ (N)')
ylabel('Count')

title('Run 27 - Load Distribution')

grid on
```

```

%% Check Load Bands

FZ = abs(data.FZ);

fprintf('\n===== LOAD BANDS =====\n\n')

fprintf('Mean Load = %.1f N\n',mean(FZ))
fprintf('Median Load = %.1f N\n',median(FZ))
%% Points in Each Load Band

FZ = abs(data.FZ);

load200 = sum(FZ > 150 & FZ < 250);
load450 = sum(FZ > 400 & FZ < 500);
load650 = sum(FZ > 600 & FZ < 700);
load900 = sum(FZ > 850 & FZ < 950);
load1100 = sum(FZ > 1050 & FZ < 1150);

fprintf('\n===== POINTS IN EACH LOAD BAND =====\n\n')

fprintf('200 N Band = %d\n',load200)
fprintf('450 N Band = %d\n',load450)
fprintf('650 N Band = %d\n',load650)
fprintf('900 N Band = %d\n',load900)
fprintf('1100 N Band = %d\n',load1100)
%% Tire Curve at 650 N

FZ = abs(data.FZ);

idx = FZ > 600 & FZ < 700;

SA = data.SA(idx);
FY = data.FY(idx);

figure

plot(SA,FY, '. ')

xlabel('Slip Angle (deg)')
ylabel('Lateral Force FY (N)')

title('Run 27 - 650 N Load Band')

grid on
%% Peak Force Extraction

[maxFY,index] = max(abs(FY));

peakSA = SA(index);

peakFZ = FZ(idx);
peakFZ = peakFZ(index);

```

```

muPeak = maxFY/peakFZ;

fprintf('\n===== PEAK TIRE PERFORMANCE =====\n\n')

fprintf('Peak FY = %.1f N\n',maxFY)
fprintf('Peak SA = %.2f deg\n',peakSA)
fprintf('Peak FZ = %.1f N\n',peakFZ)
fprintf('Peak Mu = %.3f\n',muPeak)
%% Camber at Peak Force

IA = data.IA(idx);

peakIA = IA(index);

fprintf('Peak IA = %.2f deg\n',peakIA)
%% Peak Point Verification

figure

scatter(SA,FY,'.')

hold on

plot(peakSA,...
     FY(index),...
     'ro',...
     'MarkerSize',10,...
     'LineWidth',2)

xlabel('Slip Angle (deg)')
ylabel('Lateral Force FY (N)')

title('650 N Load Band with Peak Point')

grid on
%% Load Sensitivity

FZ = abs(data.FZ);

loadBands = [200 450 650 900 1100];

peakFY = zeros(size(loadBands));
peakFZ = zeros(size(loadBands));
peakMu = zeros(size(loadBands));
peakSA = zeros(size(loadBands));

%% Extract Peak Values From Each Band
for i = 1:length(loadBands)

    load = loadBands(i);

    idx = FZ > load-50 & FZ < load+50;

    SA_band = data.SA(idx);
    FY_band = data.FY(idx);
    FZ_band = FZ(idx);

    [maxFY,ind] = max(abs(FY_band));

    peakFY(i) = maxFY;

```

```

        peakFZ(i) = FZ_band(ind);
        peakMu(i) = maxFY/FZ_band(ind);
        peakSA(i) = SA_band(ind);

end
fprintf('\n===== LOAD SENSITIVITY RESULTS =====\n\n')

for i = 1:length(loadBands)

    fprintf('Load Band = %4d N | Peak FY = %7.1f N | Peak Mu = %.3f | Peak SA = %.2f deg\n',...
        loadBands(i),...
        peakFY(i),...
        peakMu(i),...
        peakSA(i));

end
%% Peak FY vs Load

figure

plot(peakFZ,peakFY,'o-','LineWidth',2)

xlabel('Vertical Load FZ (N)')
ylabel('Peak Lateral Force FY (N)')

title('Peak FY vs Vertical Load')

grid on
%% Peak Mu vs Vertical Load

figure

plot(peakFZ,peakMu,'o-','LineWidth',2)

xlabel('Vertical Load FZ (N)')
ylabel('Peak Friction Coefficient \mu')

title('Peak Mu vs Vertical Load')

grid on
%% Camber Sensitivity at 650 N

FZ = abs(data.FZ);

idx650 = FZ > 600 & FZ < 700;

IA = data.IA(idx650);

fprintf('\n===== CAMBER POINTS =====\n\n')

fprintf('0 deg points = %d\n',sum(abs(IA-0) < 0.25))

fprintf('2 deg points = %d\n',sum(abs(IA-2) < 0.25))

fprintf('4 deg points = %d\n',sum(abs(IA-4) < 0.25))
%% Peak Force at Each Camber

idx0 = idx650 & abs(data.IA-0) < 0.25;
idx2 = idx650 & abs(data.IA-2) < 0.25;
idx4 = idx650 & abs(data.IA-4) < 0.25;

```



```

peakFY_0 = max(abs(data.FY(idx0)));
peakFY_2 = max(abs(data.FY(idx2)));
peakFY_4 = max(abs(data.FY(idx4)));

fprintf('\n===== CAMBER SENSITIVITY =====\n\n')

fprintf('0 deg Camber Peak FY = %.1f N\n',peakFY_0)
fprintf('2 deg Camber Peak FY = %.1f N\n',peakFY_2)
fprintf('4 deg Camber Peak FY = %.1f N\n',peakFY_4)
%% Peak FY vs Camber

camber = [0 2 4];

peakFY_camber = [peakFY_0 peakFY_2 peakFY_4];

figure

plot(camber,peakFY_camber,'o-','LineWidth',2)

xlabel('Camber (deg)')
ylabel('Peak Lateral Force FY (N)')

title('Peak FY vs Camber')

grid on
%% Cornering Stiffness - 650 N

FZ = abs(data.FZ);

idx650 = FZ > 600 & FZ < 700;

SA = data.SA(idx650);
FY = data.FY(idx650);

figure

plot(SA,FY, '.')

xlabel('Slip Angle (deg)')
ylabel('Lateral Force FY (N)')

title('Cornering Stiffness Investigation - 650 N')

grid on
%% Linear Region

idxLinear = idx650 & abs(data.SA) < 2;

SA_linear = data.SA(idxLinear);
FY_linear = data.FY(idxLinear);

figure
plot(SA_linear,FY_linear, '.')

xlabel('Slip Angle (deg)')
ylabel('Lateral Force FY (N)')
title('Linear Region - 650 N')

grid on

```

```

%% Very Small Slip Angles

idxSmall = idx650 & abs(data.SA) < 1;

figure
plot(data.SA(idxSmall),data.FY(idxSmall),'.')

xlabel('Slip Angle (deg)')
ylabel('Lateral Force FY (N)')
title('Very Small Slip Angles')

grid on
%% Cornering Stiffness Calculation

idxSmall = idx650 & abs(data.SA) < 1;

SA_small = data.SA(idxSmall);
FY_small = data.FY(idxSmall);

p = polyfit(SA_small,FY_small,1);

corneringStiffness = abs(p(1));

fprintf('\n==== CORNERING STIFFNESS =====\n\n')
fprintf('Cornering Stiffness = %.1f N/deg\n',corneringStiffness)

figure

plot(SA_small,FY_small,'.')
hold on

SA_fit = linspace(min(SA_small),max(SA_small),100);

FY_fit = polyval(p,SA_fit);

plot(SA_fit,FY_fit,'r','LineWidth',2)

xlabel('Slip Angle (deg)')
ylabel('Lateral Force FY (N)')
title('Cornering Stiffness Fit - 650 N')

legend('Data','Linear Fit')

grid on
%% Cornering Stiffness vs Load

loadBands = [250 450 650 900 1100];

corneringStiffness = zeros(size(loadBands));

for i = 1:length(loadBands)

    targetLoad = loadBands(i);

    idxLoad = abs(abs(data.FZ) - targetLoad) < 50;

    idxLinear = idxLoad & abs(data.SA) < 1;

    SA_temp = data.SA(idxLinear);
    FY_temp = data.FY(idxLinear);

```

```

    p = polyfit(SA_temp,FY_temp,1);

    corneringStiffness(i) = abs(p(1));

    fprintf('Load Band = %4.0f N | Cornering Stiffness = %.1f N/deg\n',...
        targetLoad,corneringStiffness(i))

end

figure

plot(loadBands,corneringStiffness,'o-','LineWidth',2)

xlabel('Vertical Load FZ (N)')
ylabel('Cornering Stiffness (N/deg)')

title('Cornering Stiffness vs Vertical Load')

grid on
%% MZ vs Slip Angle

FZ = abs(data.FZ);

idx650 = FZ > 600 & FZ < 700;

SA = data.SA(idx650);
MZ = data.MZ(idx650);

figure

plot(SA,MZ,'.')

xlabel('Slip Angle (deg)')
ylabel('Aligning Moment MZ (Nm)')

title('MZ vs Slip Angle - 650 N')

grid on
%% Peak MZ Extraction

[maxMZ,indexMZ] = max(abs(MZ));

peakMZ = MZ(indexMZ);

peakSA_MZ = SA(indexMZ);

peakFZ_MZ = abs(data.FZ(idx650));
peakFZ_MZ = peakFZ_MZ(indexMZ);

peakIA_MZ = data.IA(idx650);
peakIA_MZ = peakIA_MZ(indexMZ);

fprintf('\n===== PEAK MZ =====\n\n')

fprintf('Peak MZ = %.2f Nm\n',peakMZ)
fprintf('Peak SA = %.2f deg\n',peakSA_MZ)
fprintf('Peak FZ = %.1f N\n',peakFZ_MZ)
fprintf('Peak IA = %.2f deg\n',peakIA_MZ)
%% Peak MZ (Useful Region)

idxUseful = idx650 & abs(data.SA) < 10;

SA_useful = data.SA(idxUseful);
MZ_useful = data.MZ(idxUseful);

[maxMZ2,ind2] = max(abs(MZ_useful));

fprintf('\n===== PEAK MZ (USEFUL REGION) =====\n\n')
fprintf('Peak MZ = %.2f Nm\n',MZ_useful(ind2))
fprintf('Peak SA = %.2f deg\n',SA_useful(ind2))

```

Appendix B – MATLAB Command Window Outputs

```
ans =  
5x17 table  
  
    ET      V      SA      IA      RL      FX      FY      FZ      MX      MY      MZ      P      N      RE      TSTI      TSTC      TSTO  
    ---    ---    ---    ---    ---    ---    ---    ---    ---    ---    ---    ---    ---    ---    ---    ---    ---  
267.58  40.32   0.04  -0.027  21.381  -71.6  -77.23  -1110.4   0.96   11.6   8.49  83.94  483.85  22.108  52.45  49.06  51.53  
267.59  40.32   0.076  -0.036  21.385  -81.55  -108.42  -1110.5   0.08  13.94  5.97  83.79  483.65  22.117  52.44  49.03  51.51  
267.6   40.32   0.117  -0.034  21.382  -83.71  -140.36  -1120.5  -0.68  14.53  5.13  83.49  483.51  22.126  52.34  49.04  51.45  
267.61  40.3   0.155  -0.037  21.383  -79.37  -169.43  -1122    0.56  13.78  5.92  83.49  483.44  22.116  52.42  49.03  51.4  
267.62  40.3   0.206  -0.031  21.383  -71.55  -197.84  -1116.6   1.94  12.14  7.56  83.49  483.71  22.104  52.37  49.01  51.36  
  
ans =  
1x17 cell array  
Columns 1 through 12  
  
    {'ET'}    {'V'}    {'SA'}    {'IA'}    {'RL'}    {'FX'}    {'FY'}    {'FZ'}    {'MX'}    {'MY'}    {'MZ'}    {'P'}  
Columns 13 through 17  
  
    {'N'}    {'RE'}    {'TSTI'}    {'TSTC'}    {'TSTO'}
```

===== BASIC STATISTICS =====

```
Minimum SA = -13.39 deg  
Maximum SA = 13.16 deg  
Minimum IA = -0.05 deg  
Maximum IA = 3.97 deg  
Minimum FZ = 167 N  
Maximum FZ = 1165 N
```

===== LOAD BANDS =====

```
Mean Load = 674.6 N  
Median Load = 664.2 N
```

===== POINTS IN EACH LOAD BAND =====

```
200 N Band = 6351  
450 N Band = 6330  
650 N Band = 6260  
900 N Band = 5807  
1100 N Band = 7412
```

===== PEAK TIRE PERFORMANCE =====

```
Peak FY = 1840.2 N  
Peak SA = 10.37 deg  
Peak FZ = 681.8 N  
Peak Mu = 2.699  
Peak IA = 1.96 deg
```

===== LOAD SENSITIVITY RESULTS =====

```
Load Band = 200 N | Peak FY = 682.1 N | Peak Mu = 2.744 | Peak SA = 12.53 deg  
Load Band = 450 N | Peak FY = 1272.1 N | Peak Mu = 2.745 | Peak SA = 12.08 deg  
Load Band = 650 N | Peak FY = 1840.2 N | Peak Mu = 2.699 | Peak SA = 10.37 deg  
Load Band = 900 N | Peak FY = 2341.3 N | Peak Mu = 2.557 | Peak SA = 10.65 deg  
Load Band = 1100 N | Peak FY = 2857.1 N | Peak Mu = 2.499 | Peak SA = 10.66 deg
```

===== CAMBER POINTS =====

```
0 deg points = 3740  
2 deg points = 1266  
4 deg points = 1254
```

===== CAMBER SENSITIVITY =====

```
0 deg Camber Peak FY = 1797.3 N  
2 deg Camber Peak FY = 1840.2 N  
4 deg Camber Peak FY = 1774.2 N
```

===== CORNERING STIFFNESS =====

```
Cornering Stiffness = 518.8 N/deg  
Load Band = 250 N | Cornering Stiffness = 250.1 N/deg  
Load Band = 450 N | Cornering Stiffness = 393.8 N/deg  
Load Band = 650 N | Cornering Stiffness = 518.8 N/deg  
Load Band = 900 N | Cornering Stiffness = 619.5 N/deg  
Load Band = 1100 N | Cornering Stiffness = 678.4 N/deg
```

===== PEAK MZ =====

```
Peak MZ = 64.01 Nm  
Peak SA = -13.11 deg  
Peak FZ = 658.6 N  
Peak IA = 3.96 deg
```

===== PEAK MZ (USEFUL REGION) =====

```
Peak MZ = 42.28 Nm  
Peak SA = 2.58 deg
```